Machine Learning Interview Questions and Answers

Simple answers you can say in your own words
AI / ML Engineer interviews

How to use this: Read each question, answer out loud first, then compare with the answer here. The 'Say in interview' box is a short version you can speak in 20 to 30 seconds. These are practice questions based on common topics, not leaked questions from any company.

Part 1: Basics

Q1. What is machine learning? How is it different from AI and deep learning?

Answer:

Machine learning is when a computer learns patterns from data, instead of us writing the rules by hand. We give it many examples, and it learns to make predictions on new data.

- **AI:** the big idea of making machines act smart.
- **Machine learning:** a part of AI, where the machine learns from data.
- **Deep learning:** a part of machine learning that uses big neural networks.

Example: Spam filter: instead of writing rules for spam words, we show thousands of spam and normal emails, and the model learns the difference by itself.

Say in interview: Machine learning means the machine learns patterns from data. AI is the big field, ML is a part of it, and deep learning is a part of ML that uses neural networks.

Q2. What are the types of machine learning?

Answer:

There are three main types:

- **Supervised:** data has the correct answers (labels). The model learns to predict them. Example: predict house price, or spam or not spam.
- **Unsupervised:** no labels. The model finds patterns by itself. Example: group customers into segments.
- **Reinforcement:** an agent learns by trying actions and getting rewards or penalties. Example: a game-playing agent or a robot.

Say in interview: Supervised learns from labeled data, unsupervised finds patterns without labels, and reinforcement learns from rewards.

Q3. What is the difference between classification and regression?

Answer:

Both are supervised learning. In classification, the output is a category, like spam or not spam, or cat, dog, or bird. In regression, the output is a number, like a house price or tomorrow's temperature.

Say in interview: Classification predicts a category and regression predicts a number.

Q4. What are features and labels? What is feature engineering?

Answer:

Features are the inputs the model uses, like age, salary, and city. The label is the answer we want to predict, like 'will buy' or 'will not buy'. Feature engineering means creating better inputs from the raw data, because good features often help more than a complex model.

Example: From a date column we can create new features like day of the week, month, and is-weekend.

Say in interview: Features are the inputs and the label is the answer. Feature engineering is making better inputs from raw data, and it often improves the model a lot.

Q5. Why do we split data into train, validation, and test sets?

Answer:

We want to know how the model will work on new data it has never seen.

- **Train set:** the model learns from this (for example 70 percent).
- **Validation set:** we use it to tune the settings and compare models.
- **Test set:** used only once at the end, for the final honest score.

Say in interview: The train set is for learning, the validation set is for tuning, and the test set is for the final check on unseen data.

Part 2: Core Concepts

Q6. What is overfitting and underfitting? How do you fix them?

Answer:

Overfitting: the model learns the training data too well, even the noise, so it does very well on training data but badly on new data. It is like a student who memorizes the answers but does not understand.

- **Underfitting:** the model is too simple and does badly on both training and new data.
- **How to spot overfitting:** high training score but low test score.
- **Fix overfitting:** get more data, use a simpler model, use regularization, use cross-validation, use dropout or early stopping for neural networks.
- **Fix underfitting:** use a more powerful model, add better features, train longer.

Say in interview: Overfitting is when the model memorizes training data and fails on new data. I fix it with more data, a simpler model, or regularization.

Q7. What is the bias-variance tradeoff?

Answer:

Bias is the error from a model that is too simple (underfitting). Variance is the error from a model that is too sensitive to the training data (overfitting). A simple model has high bias and low variance. A very complex model has low bias and high variance. We want the balance in the middle, where the total error on new data is lowest.

Say in interview: High bias means too simple and underfit. High variance means too complex and overfit. We look for the balance between the two.

Q8. What is cross-validation?

Answer:

Instead of one train and test split, we split the data into k parts (for example 5). We train on 4 parts and test on 1, and repeat so that every part is used as the test once. Then we average the scores. It gives a more reliable estimate, especially when we have less data.

Say in interview: Cross-validation tests the model on different parts of the data in turns and averages the scores, so the result is more reliable.

Q9. What is regularization? What are L1 and L2?

Answer:

Regularization adds a penalty for large weights in the model, so the model stays simple and does not overfit.

- **L1 (Lasso):** can push some weights to exactly zero, so it removes useless features.
- **L2 (Ridge):** makes all the weights small but not zero.

Say in interview: Regularization penalizes large weights to avoid overfitting. L1 removes useless features, and L2 shrinks all weights.

Q10. What is gradient descent? What is the learning rate?

Answer:

Gradient descent is how a model learns. It starts with random weights, checks the error, and changes the weights a little bit in the direction that reduces the error. It repeats this many times. The learning rate is the size of each step.

- Learning rate too big: the model jumps around and may never settle.
- Learning rate too small: learning is very slow.

Example: Think of walking down a hill in fog. You feel the slope under your feet and take a step downward again and again.

Say in interview: Gradient descent reduces the error step by step by adjusting the weights. The learning rate is the step size.

Q11. What is a loss function?

Answer:

A loss function measures how wrong the model's predictions are. The model tries to make this number as small as possible during training. For regression we commonly use mean squared error. For classification we commonly use cross-entropy.

Say in interview: The loss function is the score of how wrong the model is, and training tries to reduce it.

Q12. What is data leakage?

Answer:

Data leakage is when information from outside the training data, especially from the test data or from the future, goes into the model training. The score looks great, but the model fails in real life.

- Scaling or filling missing values using the full data before splitting.
- Using a feature that is only known after the event, like using 'refund issued' to predict fraud.

- Same record present in both train and test sets.

Say in interview: Data leakage is when the model sees information it would not have in real life. The test score looks too good, but real performance is bad. I always split first, then process.

Part 3: Algorithms

Q13. How does linear regression work?

Answer:

It fits a straight line (or a flat plane for many features) through the data to predict a number. The model learns a weight for each feature, and the prediction is the sum of feature times weight plus a base value. It finds the line that has the smallest total error. It is simple and easy to explain, but it only captures straight-line relations.

Example: House price = weight1 times size + weight2 times number of rooms + base value.

Say in interview: Linear regression fits a straight line through the data to predict a number. It is simple and easy to explain.

Q14. How does logistic regression work? Why is it used for classification?

Answer:

Despite the name, it is a classification method. It works like linear regression, but then passes the result through a sigmoid function, which squeezes it into a number between 0 and 1. We read this number as a probability, and if it is above a threshold like 0.5, we say yes.

Say in interview: Logistic regression gives the probability of a class using a sigmoid function. It is simple, fast, and a good first model for classification.

Q15. How does a decision tree work?

Answer:

It asks a series of yes or no questions about the features, like 'is age more than 30?', and splits the data again and again until the groups are mostly of one class. It chooses the best question at each step using a measure like Gini or entropy. It is easy to understand, but a deep tree can easily overfit.

Say in interview: A decision tree splits the data with yes or no questions. It is easy to explain, but it can overfit if it is too deep.

Q16. What is a random forest? What is bagging?

Answer:

A random forest is many decision trees, each trained on a random sample of the data and a random set of features. For the final answer, they vote (classification) or average (regression). This idea of training many models on different samples and combining them is called bagging. It reduces overfitting and is more stable than a single tree.

Say in interview: A random forest is many random trees voting together. Combining many models reduces overfitting and gives a stable result.

Q17. What is boosting? What is the difference between bagging and boosting?

Answer:

In boosting, the models are built one after another, and each new model focuses on fixing the mistakes of the previous ones. Examples: Gradient Boosting, XGBoost, and LightGBM. They are often the best choice for table (structured) data.

- **Bagging:** models are trained in parallel and independently, then combined. It reduces variance (overfitting).
- **Boosting:** models are trained one after another, each fixing the previous errors. It reduces bias, and can overfit if not tuned.

Say in interview: Bagging trains many models in parallel and averages them. Boosting trains models one after another, each fixing the earlier mistakes.

Q18. How does KNN (K-Nearest Neighbors) work?

Answer:

To predict for a new point, it finds the k closest points in the training data, and takes the majority class (or the average value). There is no real training, but prediction is slow on big data, and it needs the features to be scaled.

Say in interview: KNN looks at the k nearest points and takes their majority vote. It is simple, but slow on big data and needs scaling.

Q19. What is SVM (Support Vector Machine)?

Answer:

SVM finds the line (or plane) that separates the classes with the widest possible gap between them. The points closest to the line are called support vectors. With a kernel, it can also separate data that is not separable by a straight line. It works well on smaller datasets with many features.

Say in interview: SVM finds the boundary with the widest gap between classes. It works well on small and medium data.

Q20. How does K-means clustering work? How do you choose k?

Answer:

K-means is unsupervised. It groups similar data points into k groups.

- Pick k random centers.
- Assign each point to the nearest center.
- Move each center to the average of its points.
- Repeat until the centers stop moving.
- To choose k, use the elbow method (plot the error against k and look for the bend) or the silhouette score.

Say in interview: K-means groups data into k clusters by repeatedly assigning points to the nearest center and moving the centers. I choose k with the elbow method.

Part 4: Metrics and Data

Q21. What are the confusion matrix, precision, recall, and F1 score?

Answer:

The confusion matrix counts the right and wrong predictions: true positives, false positives, true negatives, and false negatives.

- **Precision:** of all the ones we said positive, how many were really positive. Use it when a false alarm is costly.
- **Recall:** of all the real positives, how many did we find. Use it when missing one is costly, like cancer or fraud.
- **F1 score:** one number that balances precision and recall.

Example: Cancer test: high recall is important, because missing a sick person is very bad. Spam filter: high precision is important, because sending a good email to spam is bad.

Say in interview: Precision is how many of our positive calls were right. Recall is how many of the real positives we found. F1 balances the two.

Q22. Why is accuracy not enough? How do you handle imbalanced data?

Answer:

If only 1 percent of transactions are fraud, a model that says 'not fraud' for everything gets 99 percent accuracy, but it is useless. So for imbalanced data we use precision, recall, F1, or AUC.

- Resample: oversample the small class (like SMOTE) or undersample the big class.
- Use class weights, so mistakes on the small class cost more.
- Change the decision threshold.
- Collect more data of the small class if possible.

Say in interview: With imbalanced data, accuracy is misleading. I use precision, recall, and F1, and I fix the imbalance with resampling or class weights.

Q23. What is ROC-AUC?

Answer:

The ROC curve shows how well the model separates the two classes at all thresholds. AUC is the area under that curve. 1.0 is perfect and 0.5 is the same as random guessing. It tells how good the model is at ranking positives above negatives, and it does not depend on one threshold.

Say in interview: AUC shows how well the model separates the classes. 1 is perfect and 0.5 is random.

Q24. How do you handle missing values, outliers, and scaling?

Answer:

Real data is always messy.

- **Missing values:** remove the rows if few, or fill with the mean, median, or most common value, or use a model to predict them. Sometimes add a flag column saying the value was missing.
- **Outliers:** check if it is an error or a real rare case. Remove, cap, or transform the value. Tree models are not affected much.
- **Scaling:** bring the features to a similar range (standardization or min-max). It is needed for KNN, SVM, linear models, and neural networks. It is not needed for tree models.

Say in interview: I fill or remove missing values, check the outliers, and scale the features for distance-based or gradient-based models.

Q25. What is dimensionality reduction and PCA?

Answer:

When data has too many features, it becomes slow, noisy, and easy to overfit. Dimensionality reduction reduces the number of features while keeping the important information. PCA is the most common method. It creates new features (components) that capture the most variation in the data, so we can keep only the first few.

Say in interview: PCA reduces many features to a few new ones that keep most of the information. It makes the model faster and reduces noise.

Part 5: Neural Networks and Production

Q26. What is a neural network? What are activation functions and backpropagation?

Answer:

A neural network is layers of small units (neurons). Each neuron takes inputs, multiplies by weights, adds them, and passes the result to the next layer. Many layers can learn very complex patterns.

- **Activation function:** adds non-linearity so the network can learn complex patterns, not only straight lines. Common ones: ReLU, sigmoid, softmax.
- **Backpropagation:** after a prediction, we calculate the error and send it backward through the layers to find how much each weight caused it. Then gradient descent updates the weights.

Say in interview: A neural network is layers of neurons that learn weights. Activation functions give non-linearity, and backpropagation sends the error back to update the weights.

Q27. How do you reduce overfitting in neural networks?

Answer:

Some common ways:

- **Dropout:** randomly switch off some neurons during training, so the network does not depend on a few neurons.
- **Early stopping:** stop training when the validation error starts to go up.
- **Regularization** (L2) and **data augmentation** (making more training data from the existing data).
- Get more data, or use a smaller network.

Say in interview: I use dropout, early stopping, regularization, more data, and data augmentation.

Q28. What is model drift? How do you monitor a model in production?

Answer:

After deployment, the real world changes, and the model slowly gets worse. This is called drift.

- **Data drift:** the input data changes, for example new types of customers.
- **Concept drift:** the relation between input and output changes, for example fraud patterns change.
- Monitor the input data, the predictions, and the accuracy (when the true labels arrive), and set alerts.

- Retrain on new data when the performance drops.

Say in interview: Drift is when the real data changes and the model gets worse over time. I monitor the data and the accuracy, and retrain when needed.

If you do not know an answer, say honestly: 'I have not worked on this directly, but my understanding is...' and then explain what you know. It is better than guessing.